

Half hole and full hole dowel embedment Strength: A review of international developments and recommendations for Australian softwoods

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ABSTRACT

Embedment strength is an essential system property of the European Yield Model which is commonly used to predict the ductile capacity of timber joints using dowel-type fasteners. This paper presents a comprehensive review of international embedment test methods and prediction formulae in design codes and literature, as well as an experimental study to evaluate the impact of different test methods parallel to the grain. A total of 224 full hole (FH) tension and half hole (HH) compression tests were conducted according to AS/NZS ISO 10984.2 on Australian sawn softwoods (Radiata Pine and Southern Pine) to compare embedment strength resulting from FH specimens sized to EN 383 and HH specimens sized to ASTM D5764, respectively. The results were compared to dowel embedment strength formulae from literature, those contained in Eurocode 5, the National Design Specification (NDS) for Wood Construction, and the respective joint group strength values contained in AS 1720.1, which are based on empirical test data. It was found that all strength prediction models were non-conservative except for a formula proposed by Smith et al. in 2004, which was based on FH tension tests in Australian Radiata Pine and Slash Pine. Furthermore, a strong correlation between timber density and embedment strength was found for both test methods, however, only FH tests displayed a correlation between dowel diameter and embedment strength, which is consistent with the embedment strengths formulae currently contained in Eurocode 5 and the NDS.

1. Introduction

The performance of timber joints directly affects a timber structure's static and dynamic response, structural appearance, and member size [1]. Timber joints are often detailed to provide ductility and energy dissipation in case of overloading and accurately predicting joint capacity is crucial for capacity design of timber structures [2]. Metal dowel-type fasteners such as dowels, bolts, nails and screws are commonly used in both light timber framed and mass timber construction. Internationally, the European Yield Model (EYM) based on the Johansen's Equations [3] is widely accepted to predict ductile joint capacity. The EYM can be derived from static equilibrium (or virtual work), assuming elastic-plastic material behaviour in the metal fasteners and wood in which the fastener is embedded. The embedding is characterised by the timber embedment strength $f_{h,i}$, using the projection of the contact surface. The fastener is assumed to develop plastic hinges, characterised by the fastener yield moment M_y .

While there is some debate about the appropriate value of the

fastener yield moment [4–6], it can be determined with relative accuracy due to low variability of material properties. M_y is either calculated using the fastener's tensile strength and geometry, or established experimentally in bending [7,8].

Similarly, semi-empirical models are available to predict fastener embedment strength in softwood and hardwood at 12% equilibrium moisture content (EMC) and other EMCs [9–21]. It should be noted that there are also several studies on embedment strength of engineered wood products, including plywood, Glulam, LVL, and CLT, but these are not explicitly covered here. Furthermore, the focus is on smooth larger diameter fasteners (bolts and dowels); screws, nails, or threaded rods are not explicitly considered.

Several variations and simplifications of the EYM have been developed over time [22–25] and adopted in different international design standards, including the European Eurocode 5 [26], the US National Design Specification (NDS) for Wood Construction [27], the Canadian Engineering Design in Wood Standard CSA-O86 [28], the Brazilian Timber Structures Design Code ABNT NBR 7190 [29] and Chinese

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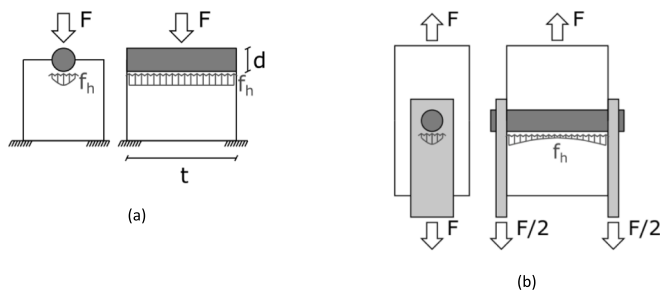


Fig. 1. a) Half hole compression and b) full hole tension embedment test and resulting embedment stress distributions. Note, embedment strength is calculated with the projected area as $f_h = F/(td)$.

National Standard for Design of Timber Structures GB 50005 [30].

However, both the Australian Standard AS 1720.1–2010 [31] and New Zealand Standard NZS 3603:1993 [32] use connection design approaches based on empirical tabulated test data [1,18,33]. Extensive experimental testing has been conducted in Australia to validate the EYM for single fastener timber joints to support adoption of limit states design [1,34–37]. Yet, apart from Smith et al. [33], little dedicated embedment testing has been conducted on Australian timber products to verify international embedment strength formulae [19]. In order to transition the Australian and New Zealand timber design standards towards the EYM, it is crucial to also ensure the validity and safety of available embedment strength formulae.

1.1. Embedment strength test methods and strength predictions

The two main test setups used to establish embedment strength are half-hole (HH) and full-hole (FH) tests, as shown in Fig. 1. HH tests are conducted in compression and ensure an even stress distribution under the fastener surface, whereas FH tests are effectively a single fastener joint with two steel side plates conducted either in compression or tension, resulting in a parabolic stress distribution [38].

These two distinct test setups are based on studies by Whale and Smith [10] and Wilkinson [12] and numerous embedment strength prediction formulae have been developed since. Table A1 in the Appendix gives a comprehensive chronological list of embedment studies on smooth bolts and dowels and suggested embedment strength prediction formulae published between 1930 and 2022 [9–21,23,26–29,33,35,38–74]. The focus is on sawn timber; engineered wood products are not considered. Embedment tests on screws, nails, or threaded rods are omitted.

The following section outlines the evolution of international embedment test standards and provides a review of studies that have evaluated the influence of different methods on embedment test results.

Between 1986 and 1988, Whale and Smith [10,69,75] carried out FH tension and compression tests on nails and bolts, which later led to the European test method to determine embedment strength and foundation values for dowel type fasteners EN 383 [53,55,64].

In 1991, Wilkinson [12] carried out HH compression tests on bolts and nails for the United States Department of Agriculture (USDA). The bolt embedment study comprised 240 tests on Douglas Fir, Southern Pine, Spruce-Pine-Fir, Sitka Spruce, Red Oak, Yellow Poplar, and Aspen with various diameters and specimen thicknesses. Wilkinson's work laid the foundation of the US Standard Test Method for Evaluating Dowel-Bearing Strength of Wood and Wood-Based Products ASTM D5764-95 [48].

According to Sawata and Yasumura [16], Hirai [76] first reviewed the influence of the embedment test method and fastener diameter on the embedment strength and stiffness. According to Awaludin et al. [45], Hirai [76] carried out tests following the experimental setups proposed by Whale and Smith [10] and Wilkinson [12]. Hirai [76] concluded that the results were only comparable for loading parallel to

the grain.

While not comparing the different test methods, Sawata and Yasumura [16] compared different methods to derive embedment strength; the 5%-offset method to evaluate yield embedment strength, $f_{h,5\%}$, and the maximum strength within the first 5 mm displacement according to EN 383 to evaluate ultimate embedment strength, $f_{h,5mm}$. They conducted a total of 1009 FH compression tests on different grades of Japanese Ezomatsu and Todomatsu according to EN 383:1993 [55], although it appears that preloading was omitted. The embedment strength parallel to the grain, $f_{h,0}$, and perpendicular to the grain, $f_{h,90}$, were established for dowel diameters of $d = 8, 12, 16, 20$ mm. They found that 5%-offset embedment strengths ($f_{h,0,5\%}$ and $f_{h,90,5\%}$) were independent of the dowel diameter, d , and that $f_{h,0,5\%}$ and $f_{h,0,5mm}$ were almost identical ($f_{h,0,5\%} / f_{h,0,5mm} = 0.968$ mean). However, $f_{h,90,5mm}$ was inversely correlated with the dowel diameter, such that $f_{h,90,5\%} / f_{h,90,5mm} = 0.016 d + 0.555$. All embedment strengths were positively correlated with density.

Santos et al. [77] reviewed the difference between EN 383:1993 [55] and ASTM D5764-97a [68]. They conducted a total of 200 embedment tests on 30 mm thick Maritime Pine wood specimens, with either 24 or 26 replicas per series. Both HH and FH specimens were loaded parallel and perpendicular to the grain in compression with a 14 mm diameter dowel. ASTM HH specimens were loaded monotonically, while EN FH specimens followed the preloading procedure before being loaded until failure. Interestingly, neither standard requires loading until failure, which provides some interesting insights. Santos et al. [77] found that the parallel to grain EN FH specimens yielded about twice the displacement at failure as the corresponding ASTM HH specimens. Similar to other studies, they found a significant correlation between wood density and embedment strength, but no correlation between density and foundation modulus (embedment stiffness). Similar to Hirai [76], Santos et al. [77] also found that ASTM HH and EN FH compression tests yielded almost identical average values parallel to grain, but 36% higher values for ASTM HH tests perpendicular to grain, while initial stiffness was similar for both orientations. In addition, they found good agreement between experimental results and embedment strength predictions in the NDS, but some deviation from Eurocode 5. However, they pointed out that the range of densities was limited, the study only used one dowel diameter, and none of the test series contained sufficient replicas to derive characteristic values.

Franke and Magnière [67] reviewed the embedment test standards EN 383:2007 [64], ASTM D5764-97a [78], and ISO 10984-2:2009 [79], and encouraged a discussion of the appropriate embedment test method at the INTER meeting in 2014 [80]. Franke and Magnière conducted FH compression embedment tests according to EN 383:2007 [64], and both HH and FH compression embedment tests according to ASTM D5764-97a [78], which prescribes different dimensions than EN 383:2007 [64]. The tests were carried out on sawn Spruce conditioned at 20 °C and 65% RH which is assumed to result in 12% EMC. The dowel diameter was $d = 12$ mm, and tests were conducted parallel and perpendicular to grain ($\alpha = 0^\circ$ and $\alpha = 90^\circ$) with 5 replicas per configuration (total of 30 tests). All specimens were preloaded according to EN 383:2007 [64] and the embedment strength was determined both at yield with the 5%-offset method as $f_{h,5\%}$, and maximum strength within the first 5 mm of displacement, $f_{h,5mm}$. There was little difference in embedment strengths perpendicular to the grain for different test methods, however, there was up to 76% difference between $f_{h,0,5\%}$ and $f_{h,0,5mm}$ due to different failure modes. For tests parallel to the grain, no difference was observed between $f_{h,0,5\%}$ and $f_{h,0,5mm}$, however, there was a significant difference between embedment strengths for different test methods. The EN FH method resulted in the lowest strength, followed by ASTM FH tests, with ASTM HH resulting in the highest strength (30% higher than EN FH). Franke and Magnière [67] recommended to use ASTM HH tests (presumably with preloading) and $f_{h,5\%}$ to determine embedment strength since this eliminates fastener bending and results in a uniform stress distribution under the dowel, which is the basis for the EYM used in

Table 1

Embedment test specimen dimensions for half hole (HH) and full hole (FH) tests and respective dimensions given in ASTM D 5764-97a [103] and EN 383:2008 [64], rounded to zero decimals and indicated in curly brackets.

	dowel diameter	hole diameter	thickness	width	loaded length	clear length	total length
	d [mm]	d_{hole} [mm]	t [mm]	w [mm]	l_1 [mm]	l_2 [mm]	l [mm]
ASTM D 5764 [103]		$d + 1.8$	$\geq \min(2d; 38)$	$\geq \max(4d; 50)$	$\geq \max(4d; 50)$	n/a	
EN 383 [64]		n/a	$1.5 \cdot 4d$	$\geq 6d$	$\geq 7d$	$\geq 30d$	$\geq 44d$
HH	12.7	13 { $\neq 14.3$ }	35 { ≥ 25.4 }	90 { ≥ 51 }	50 { $\neq 51$ }	–	50
HH	16	17 { $\neq 18.6$ }	35 { ≥ 32 }	90 { ≥ 64 }	65 { ≥ 64 }	–	65
FH	12.7	13 {n/a}	35 {19–51}	90 { ≥ 76 }	84 { $\neq 89$ }	356 { $\neq 381$ }	440
							{ $\neq 559$ }
FH	16	17 {n/a}	35 {24–64}	90 { $\neq 96$ }	112 { ≥ 112 }	588 { ≥ 480 }	700 { $\neq 704$ }

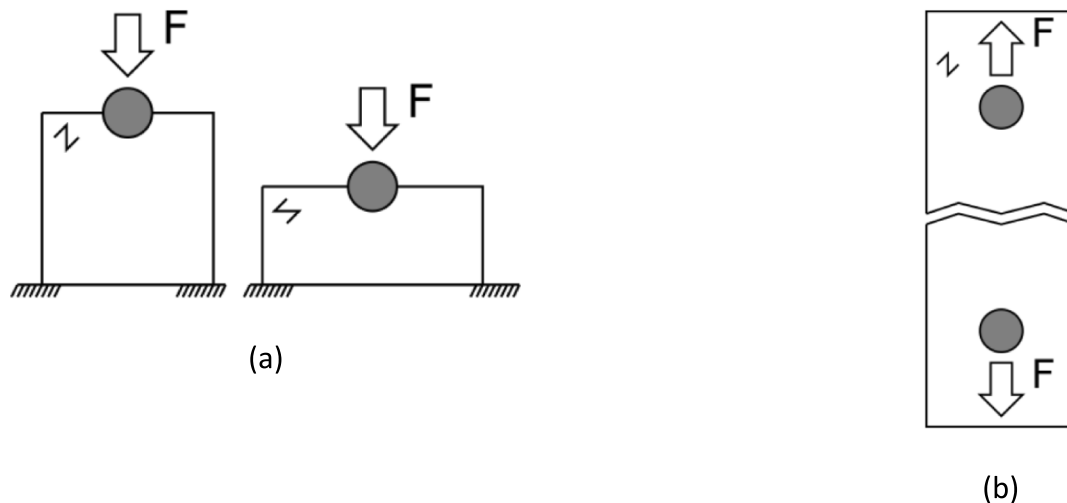


Fig. 2. a) Half hole compression and b) full hole tension tests as stipulated in AS/NZS/ISO 10984–2:2015 [93].

Eurocode 5 [26]. They also recommended a specimen size according to EN 383:2007 [64] with the additional requirement of $t \geq \min(40 \text{ mm}, 4d)$. In the discussion that followed, it was remarked that connection and embedment stiffness is rarely used in practice and it was generally agreed that the test method should be as simple as possible to eliminate human error [80]. From the discussions, there was no clear conclusion as to whether the EYM in Eurocode 5 [26] should be used with FH or HH test values and whether these should be established with the 5%-offset at yield, or within the first 5 mm displacement [80].

Currently, Eurocode 5 [26] embedment strength predictions are based on FH tests conducted by Whale and Smith [10], Whale et al. [4], Smith et al. [46], and Ehlbeck and Werner [13,52], which subsequently led to the development of EN 383 [53,70]. However, since EN FH tests result in the lowest embedment strengths according to Franke and Magnière [67], one might conclude that the embedment strength predictions currently contained in Eurocode 5 [26] are conservative, and even more so for hardwoods as highlighted in Table 1A. (That this is not the case in the present study will be demonstrated further below).

Molina et al. [71] conducted a total of 48 embedment tests on Slash Pine and Blue Gum parallel and perpendicular to the grain. The tests were FH compression tests with preloading according to EN 383:2007 [64] and the Brazilian ABNT NBR 7190:1997 Standard [29], which determines embedment strength at 2% specific deformation (strain), where specific deformation is determined by dividing the load-induced deformation by the specimen length beneath the dowel ($14d$). They concluded that the ABNT NBR 7190:1997 [29] method resulted in lower embedment strengths and recommended the use of EN 383:2007 [64] instead.

Malek et al. [72] conducted 18 HH embedment tests according to ASTM D5764-97a [65] parallel to grain on Mengkulang and Kempas using a $d = 12 \text{ mm}$ dowel. They subsequently determined $f_{h,5\%}$ and

$f_{h,5\text{mm}}$ and found that $f_{h,5\text{mm}}$ was on average 7.71% and 21.45% higher than $f_{h,5\%}$, for Menkulang and Kempas, respectively.

Yurrita and Cabrero [73,21] compared several proposed and established embedment strength formulae for FH tests, including those by Fahlbusch [9], Norén [42], Ehlbeck and Werner [13], Sawata and Yasumura [16], Leijten and Köhler [15] or rather Leijten et al. [56], Jumaat et al. [60], and Sandhaas et al. [20], as well as Eurocode 5 [26], the NDS [27], and the embedment formula for Radiata Pine for the proposed revision of NZS 3603, DZ NZS AS 1720.1 [81]. They also raised the issue that many such formulae are fitted to mean densities to obtain the mean embedment strength, which does not imply that the use of the characteristic density would predict the characteristic embedment strength accurately. Yurrita and Cabrero proposed a formula for FH embedment strength parallel to the grain based on Sandhaas et al.'s [20] dataset which is independent of the dowel diameter, but correlated to timber density (with a quadratic relationship) and implicitly considers dowel hardness through the yield strength. This formula was then validated with Leijten et al.'s [56] compiled dataset.

Schweigler et al. [74] developed a beam on elastic foundation embedment model to describe FH embedment load–displacement curves. They derived input parameters for their model based on 7 FH datasets from ENSTIB/LERMAb Epinal, Vienna University of Technology (IMWS, TU Wien), Linnaeus University Växjö (LNU), TU Delft and Karlsruhe Institute of Technology (KIT). Importantly, they recommended increasing the displacement limit of EN to 15 mm to capture the entire embedment behaviour. Furthermore, they recommended reinforcing embedment specimens sensitive to splitting, and to define specimen dimensions for loading at an angle to the grain, and amending EN 383 to include additional embedment parameters that can be used for numerical models.

It should be noted that apart from timber density, fastener diameter,

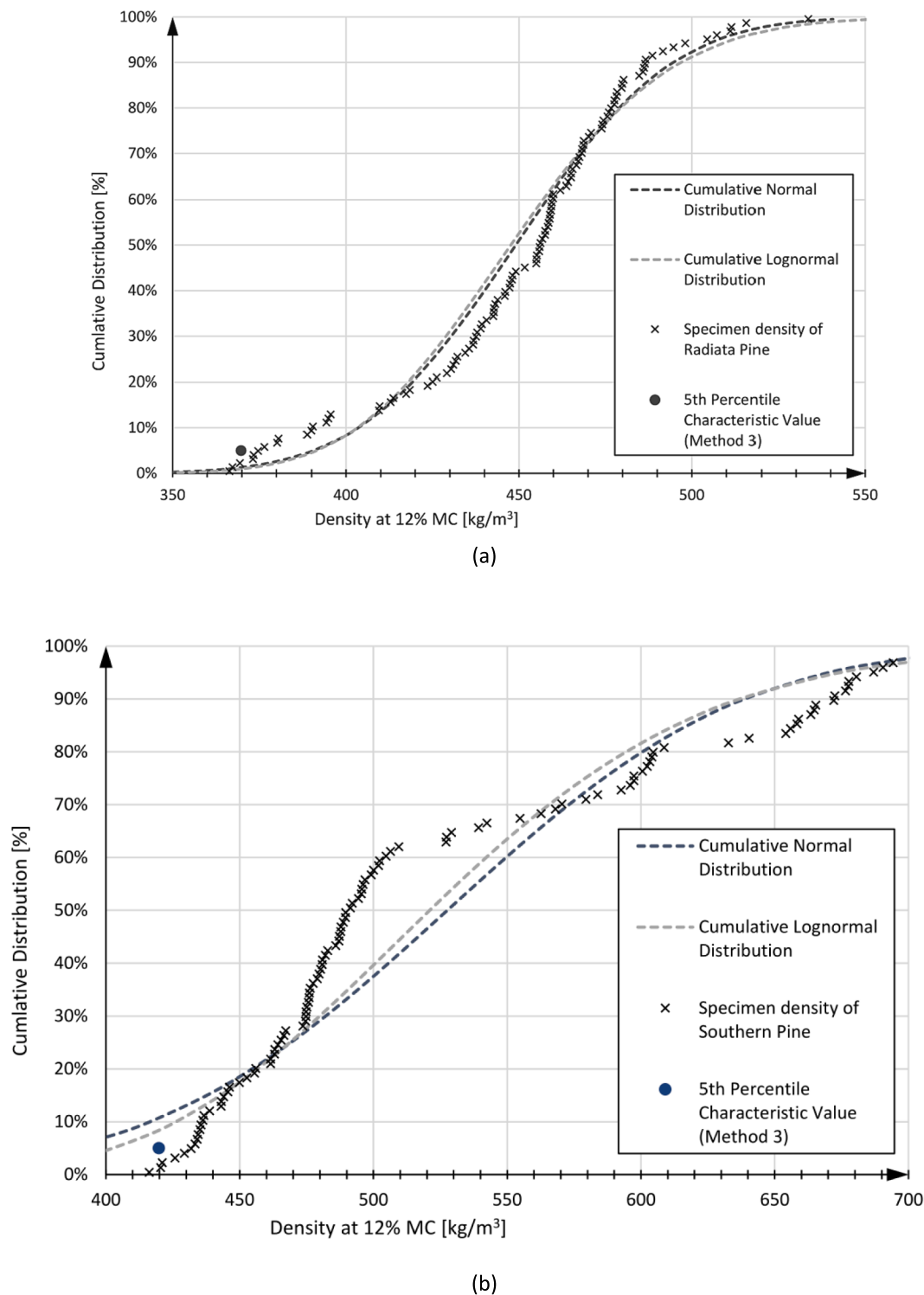


Fig. 3. Cumulative density distributions and respective characteristic density values for a) Radiata Pine and b) Southern Pine.

and load-to-grain angle, other factors may influence embedment strength, and some of these were reviewed by Smith and Foliente [82], Zhou and Guan [83], and Li [84], including fastener shape [85,22,24], fastener slenderness [86,70] fastener roughness and hardness [87–89,20,21,73], initial contact conditions [38], and wood surface conditions, including drilling damage [90]. Additional factors that have been studied include moisture content [14,91], and temperature [92].

Finally, although the joint capacity prediction models in AS 1720.1–2010 [31] are based on empirical data and do not require

embedment strength values, Australia adopted the ISO standard 10984–2 [79] for timber embedment testing. The standard was published as AS/NZS/ISO 10984–2:2015 [93] and allows for both FH and HH testing, where the FH method is carried out in tension parallel to the grain, and HH testing is conducted in compression both parallel and perpendicular to the grain, as shown in Fig. 2.

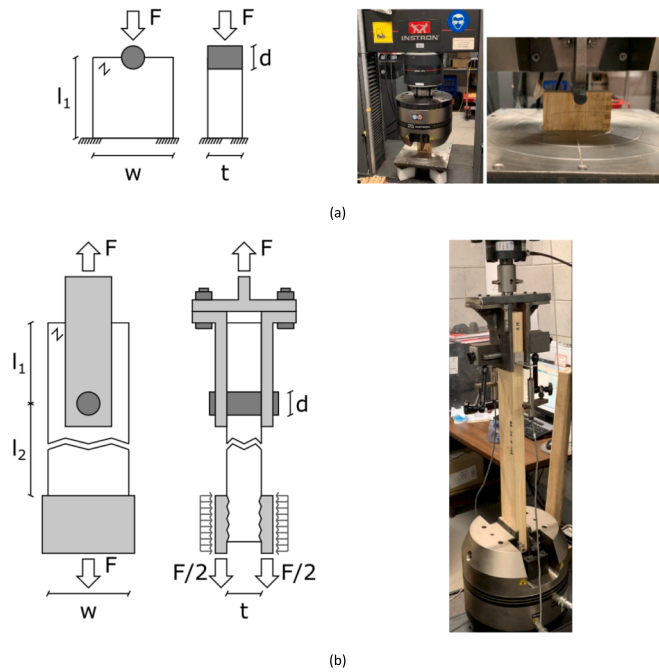


Fig. 4. a) HH compression and b) FH tension embedment test setups.

1.2. Joint capacity prediction in Australian and New Zealand standards

The strength prediction for bolted and dowelled joints in both the New Zealand Timber Design Standard NZS 3603:1993 [32] and Australian Timber Design Standard AS 1720.1–2010 [31] is based on an empirical fit of experimental data and not the EYM [1,19]. The joint capacity for a single fastener is provided as tabulated data and depends only on the fastener diameter, the timber thickness, and the timber species. While the standards employ slightly different modification factors, the design methods are similar; neither predict the yield mode, or brittle failure; and hence can lead to overestimating of the joint capacity [19].

Leicester [94] gives a summary of the early development of Australian Standards for the design of joints in timber structures, i.e. the 1988 version of AS 1720.1 [95] to determine the allowable design capacity for joints (previously published in 1972 as CA65 [96]), and AS 1649 [97], published in 1974, which gives procedures for experimental testing of joints using metal fasteners in timber. Both standards have since undergone a soft conversion to limit states design from their original working stress design format [34]. According to Leicester [94], both Australian Standards were based on North American research (such as Trayer [39]), “supplemented by Australian studies and experience related to hardwood species”, published in the first Australian timber design handbook by Langlands and Thomas in 1939 [98], and updated design guidelines for joints with metal connectors provided in the Timber Engineering Design Handbook by Pearson, Kloot, and Boyd in 1958 [99]. The timber joint design in AS 1720.1–1997 [100] was further updated based on experimental work on joints in Australian timbers carried out by the Commonwealth Scientific and Industrial Research Organization (CSIRO) by Mack [44], Lhuide [101,102], and Stringer [38]. While both softwoods and hardwoods were included, the main focus was on nailed joints as these were predominantly used in residential timber framed construction.

This empirical approach has the shortcoming that the design procedures contained in AS1720.1 [95,200] are only applicable to the fasteners and materials previously tested, and cannot be safely extrapolated to new products or timber materials [1,19,35,36]. To rectify this, work was undertaken by the Forest and Wood Products Research and Development Corporation in the early 2000s to develop a limit states

design procedure for Australian timber joints and investigate the use of the EYM in Australian timber materials [80,33,35,36]. Comprehensive experimental embedment and joint testing was undertaken on matched specimens and EYM-based equations were calibrated to Australian test data [35]. However, this work was not incorporated in AS 1720.1–2010 [31], although efforts were made to include both the EYM and brittle failure models into a draft revision for a joint New Zealand Australian Standard DZ NZS AS 1720.1 [81]. In the meantime, AS/NZS ISO 10984.2:2015 [93] was adopted in both New Zealand and Australia, as noted above. This standard allows for both FH tension embedment tests parallel to the grain, as well as HH compression embedment tests parallel and perpendicular to the grain, both with and without preloading.

2. Materials and methods

The objectives of the experimental study were 1) to establish embedment values for softwoods commonly used in Australian residential construction, 2) to assess the applicability of embedment formulae proposed in literature, and 3) to compare embedment strength obtained by full hole (HH) tension and half hole (HH) tests, both of which are stipulated in AS/NZS ISO 10984.2:2015 [93], and EN 383:2007 [64] and ASTM D5764-97a [103], respectively.

The timber was supplied by AKD Softwoods: Australian Radiata Pine and Southern Pine, a hybrid of Caribbean Pine and Slash Pine, with all timber being plantation timber from South East Queensland. The seasoned timber boards were 90 mm wide, 35 mm thick, and 2.4 m long, which is a typical Australian framing timber size.

In order to match samples, the density of each board was measured to identify suitable pairs of FH and HH test specimens. Required clear lengths were identified for FH specimens, with adjacent HH specimens where possible. However, exact matching was not always possible due to density variations, and the HH samples displayed on average higher density than FH samples for Radiata Pine, and lower density for Southern Pine. The boards were cut to size and all specimens were conditioned in an environmental chamber prior to testing. It is important to note that while the grade stamp suggested the Radiata Pine was graded MGP 10 according to AS/NZS 1748:2006 [104], this grading is no longer valid once timber has been cut to size.

The density of each test specimen was measured prior to testing, and the equivalent density at 12% EMC was computed based on oven-dry densities as $\rho_m = 467 \text{ kg/m}^3$ for Radiata Pine, and $\rho_m = 529 \text{ kg/m}^3$ for Southern Pine. Furthermore, in order to predict embedment strength with formulae from literature, the characteristic density was calculated for each species as the 5-th percentile value estimated with 75% confidence according to AS/NZS 4063.2:2010 [105] Method 3. While this non-parametric statistical evaluation method is generally used to establish characteristic strength, it was deemed appropriate in the given case since the characteristic density was employed to predict characteristic strength, the confidence coefficient was established as $k_z = 1.8$, and the minimum sample size of $n = 30$ was met (the sample size was $n = 112$ for each timber species). Using Method 3, the characteristic density ρ_k was then calculated based on ranked 5-th percentile density ρ_{05} with Equation (1), where ρ_{05} was derived from the empirical cumulative frequency distribution of density data as shown in Fig. 3.

$$\rho_k = k_s \rho_{05} \quad (1)$$

where k_s denotes the sampling factor calculated as

$$k_s = 1 - 1.8V_R/\sqrt{n} \quad (2)$$

with sample size n and coefficient of variation V_R .

This resulted in a characteristic density of $\rho_k = 370 \text{ kg/m}^3$ for Radiata Pine, and $\rho_k = 420 \text{ kg/m}^3$ for Southern Pine.

The moisture content was obtained directly after testing by means of oven-drying at 103 °C according to AS/NZS 1080:2012 [106]. The HH specimens were dried in their entirety, while smaller pieces were cut

Table 2

Comparison of Mack [44] experimental data for steel-to-timber tests (effectively FH C) and comparison to strength predictions from literature using experimental mean air-dry density ρ_m , with dowel diameter d , timber thickness t . F_{max} denotes maximum observed strength, n designates number of replicas. Strength ratios ≥ 1.0 indicate a conservative strength prediction. Unacceptably non-conservative predictions (<0.95) are printed bold.

	Hoop Pine	Loblolly Pine	Slash Pine	Caribbean Pine	Patula Pine	Spotted Gum		Radiata Pine						Average	Comment
d [mm]	12	12	12	12	12	10	20	10	10	12	12	20	20		
t [mm]	35	35	35	35	35	35	35	35	70	35	70	35	70		
n	13	14	14	14	14	10	10	10	10	10	10	10	10		
F _{max} [kN]	19.3	15.1	16.2	14.8	12.3	31.2	34.8	15.5	28.5	17	40.3	29.1	56		
ρ _m [kg/m ³]	517	529	546	470	422	1047	1047	540	540	540	540	540	540		
Mack [44]	1.20	0.92	0.95	1.03	0.96					1.01				1.01	mostly conservative
Smith et al [33]	1.46	1.12	1.16	1.23	1.14	1.35	0.86	1.30	1.19	1.23	1.46	1.40	1.35	1.25	conservative (except Spotted gum d = 20 mm)
Eurocode 5 [26]	1.23	0.94	0.98	1.04	0.96	1.15	0.72	1.11	1.02	1.04	1.23	1.17	1.13	1.06	mostly conservative
Ehlbeck & Werner [13]						0.93	0.58							0.75	non-conservative
Sawata & Yasumura [16]	1.01	0.78	0.81	0.86	0.79	0.97	0.54	0.94	0.86	0.86	1.01	0.88	0.85	0.86	non-conservative
Leijten and Köhler [15]	1.10	0.84	0.87	0.94	0.87	0.93	0.62	0.97	0.89	0.93	1.10	1.08	1.04	0.94	non-conservative
Traetta & Schickhofer [61]	1.29	0.99	1.03	1.09	1.01	1.20	0.74	1.16	1.07	1.09	1.29	1.21	1.16	1.10	conservative (except Spotted gum d = 20 mm)
Sandhaas et al. [20]	1.08	0.83	0.86	0.91	0.85	1.04	0.58	1.00	0.92	0.91	1.08	0.94	0.90	0.92	non-conservative

from the FH samples. The mean moisture contents were 10.7% and 12.3% for HH and FH samples, respectively. To eliminate effects of varying moisture contents, the equivalent density at 12% MC was employed in all strength predictions.

Smooth plain shank dowels with two different diameters (12.7 mm and 16 mm) were used for each test series. The dowels were made of high tensile steel alloy 4140 T with medium hardenability, with a 0.2% proof stress of 665 MPa, and maximum tensile stress of 850–1000 MPa, all compliant with AS 1444:2007 [107]. While these characteristics are provided for replicability purposes, the effects of fastener roughness and hardness, or initial contact and wood surface conditions were not studied.

2.1. Test setup

Fig. 4 shows the FH tension and HH compression test setups. FH and HH specimens were sized according to EN 383:2008 [62] and ASTM D5764-97a [103], respectively. The sizes of EN 383:2008 [64] and AS/NZS ISO 10984.2:2015 [93] are identical, except for the thickness being $t = 4 \dots 5 d$ in the ISO standard, which may well be an error. Adjustments were made based on available sawn timber sizes, and the clear length for FH tension tests was reduced since jaw grips were used to introduce the load. The specimen dimensions are given in Table 1.

All experiments were conducted with a 5985 Instron test machine with a load displacement rate of 2 mm/min. All samples were subjected to monotonic loading with preloading according to AS/NZS ISO 10984.2:2015 [93], however, the pre-load was reduced from 40% to 20% of the estimated maximum load to prevent premature embedment during the preloading cycle, as suggested by Ottenhaus et al. [2].

Displacements of the FH tests were measured with two linear variable differential transformers (LVDTs) installed between each end of the dowel and the timber surface according to EN 383:2008 [64]. For the HH tests, the Instron's internal deformation gauge was used to establish embedment displacement since the specimens were too short to install LVDTs.

3. Theory and analysis

The first objective of the present study was to establish embedment values for softwoods commonly used in Australian residential construction. Limited embedment testing was undertaken by Smith et al. [33], however, an earlier study by Mack [44] conducted quasi FH compression tests on several Australian softwoods and hardwoods, both seasoned and unseasoned, parallel and perpendicular to the grain. In a first step, Mack's experimental data from seasoned FH tests parallel to the grain were analysed and compared to FH strength prediction models from literature, selected based on their relevance to Mack's study from 1981 and the present experimental study. Equations 3.1 to 10 give full hole embedment strength predictions for $f_{h,5mm}$ or $f_{h,max}$ parallel to grain, which are used to calculate ultimate joint capacity. The respective literature reference is given with each Equation. Note that as stated by Yurrita and Cabrero [21], it is possible to calculate mean embedment strength with a characteristic strength prediction formula by inserting the mean density, but not vice versa.

Mack [44] (FH C):

$$F_{max} = 0.0138\rho_{12}^{1.13} \text{ (for } d = 12 \text{ mm, } t = 35 \text{ mm)} \quad (3.1)$$

$$F_y = 0.129\rho_{12}^{0.68} \text{ (for } d = 12 \text{ mm, } t = 35 \text{ mm)} \quad (3.2)$$

Smith et al. [33] (FH T):

$$f_{h,5mm,k} = 0.1\rho_k d^{-0.2} \quad (4)$$

EC 5 [26] (FH T & C) for $f_{h,5mm}$:

$$f_{h,5mm,k} = 0.082 (1 - 0.01 d)\rho_k \quad (5)$$

Ehlbeck and Werner [13] (FH T & C):

$$\begin{aligned} \text{Softwoods: } f_{h,5mm,k} &= 0.082 (1 - 0.01 d)\rho; r \\ &= 0.49 \text{ for 360 tests (same as EC 5)} \end{aligned} \quad (6.1)$$

$$\text{Hardwoods: } f_{h,5mm,k} = 0.102 (1 - 0.01 d)\rho; r = 0.68 \text{ for 119 tests} \quad (6.2)$$

Sawata and Yasumura [16] (FH C):

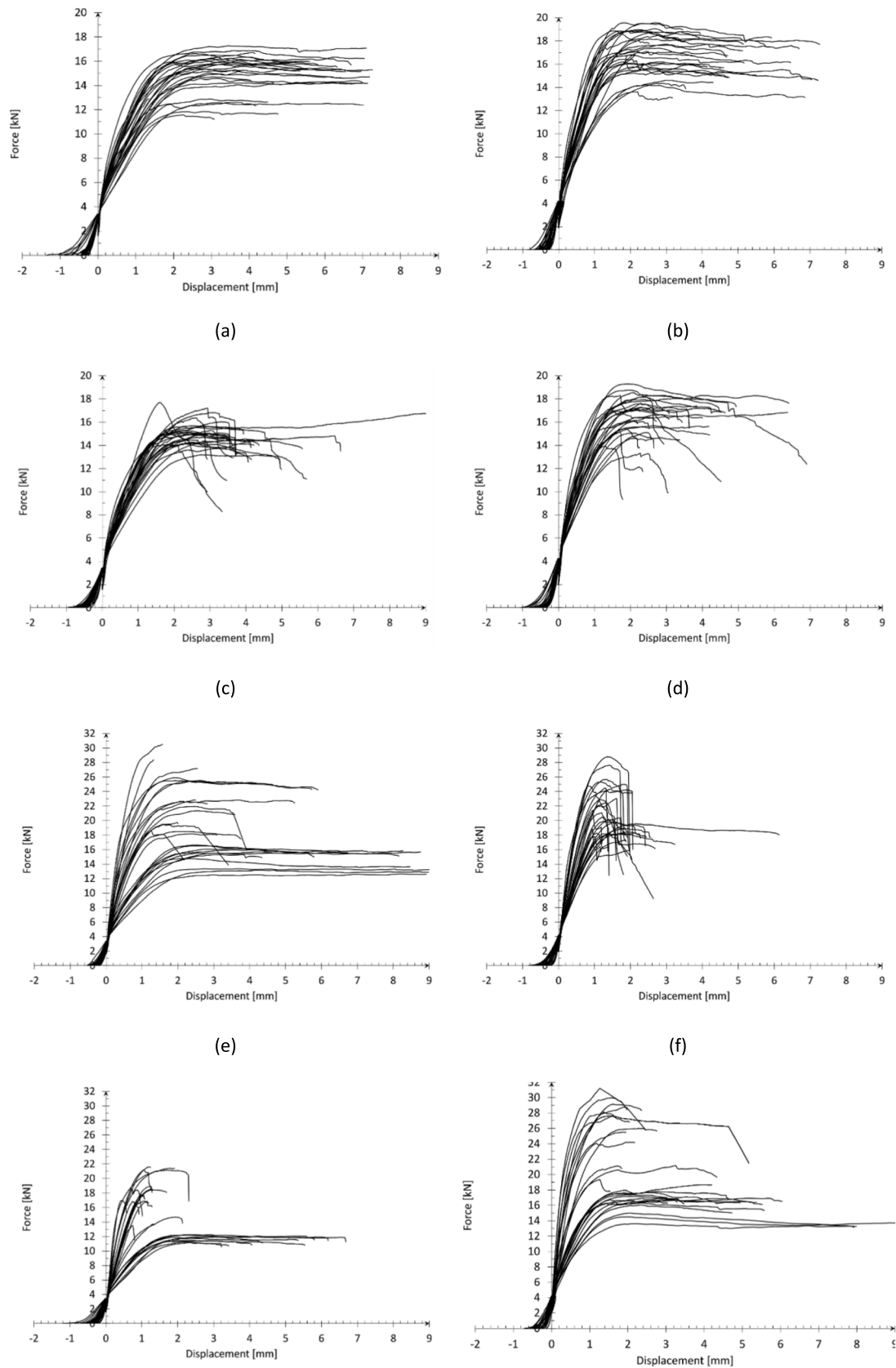


Fig. 5. Embedment load displacement curves for full hole (FH) tension and half hole (HH) compression tests. a) Radiata Pine FH $d = 12$ mm, b) Radiata Pine FH $d = 16$ mm, c) Radiata Pine HH $d = 12$ mm, d) Radiata Pine HH $d = 16$ mm, e) Southern Pine FH $d = 12$ mm, f) Southern Pine FH $d = 16$ mm, g) Southern Pine HH $d = 12$ mm, h) Southern Pine HH $d = 16$ mm.

Table 3

Results from full hole (FH) embedment tests, with respect dowel diameter, d , mean and characteristic densities adjusted for 12% MC, $\rho_{m,12}$ and $\rho_{k,12}$, mean and characteristic embedment strengths established with the 5% offset method between 10% and 40% of F_{max} , $f_{h,5\%(10-40),m}$ and $f_{h,5\%(10-40),k}$, mean and characteristic embedment strengths established with the 5% offset method between 20% and 60% of F_{max} , $f_{h,5\%(20-60),m}$ and $f_{h,5\%(20-60),k}$, mean and characteristic maximum embedment strengths within the first 5 mm after the reloading point, $f_{h,5mm,m}$ and $f_{h,5mm,k}$, and respective standard deviation indicated in brackets (SD). Sample size of $n = 28$ for each setup.

Setup	Radiata Pine FH		Southern Pine FH		Radiata Pine HH		Southern Pine HH	
d [mm]	12.7	16	12.7	16	12.7	16	12.7	16
n	28	28	28	28	28	28	28	28
$\rho_{m,12}$ [kg/m ³]	438.1 (34.1)	429.7 (33.2)	561.3 (91.6)	547.8 (91)	464.9 (25.4)	463.8 (36)	508 (88.7)	511.8 (68.3)
$\rho_{k,12}$ [kg/m ³]	366.5	367.4	433.2	420.6	390.5	373.3	418.0	442.3
MC [%]	12.9 (0.9)	12.3 (0.8)	11.3 (1.1)	12.8 (1)	11.5 (0.7)	11.2 (0.5)	10.6 (1.4)	9.6 (0.6)
$f_{h,5\%(10-40),m}$ [MPa]	29.5 (2.7)	28.9 (2.9)	41.8 (11.3)	37 (9.5)	31.1 (2.7)	28.1 (2.6)	33.8 (8.2)	37.1 (5.8)
$f_{h,5\%(10-40),k}$ [MPa]	25.3	23.7	26.0	23.8	26.0	22.5	23.1	27.7
$f_{h,5\%(20-60),m}$ [MPa]	31.4 (3.1)	29.1 (3)	42.8 (10.8)	37.2 (9.5)	31.7 (2.3)	28.3 (2.6)	34.1 (7.9)	37.2 (5.8)
$f_{h,5\%(20-60),k}$ [MPa]	25.5	23.8	27.2	24.0	27.8	22.5	24.8	27.8
$f_{h,5mm,m}$ [MPa]	33.0 (3.6)	29.4 (2.9)	44.6 (11.0)	37.7 (9.6)	33 (2.2)	28.8 (2.6)	34.6 (7.8)	37.5 (6.0)
$f_{h,5mm,k}$ [MPa]	25.5	23.9	28.3	24.3	28.7	23.1	25.3	28.4

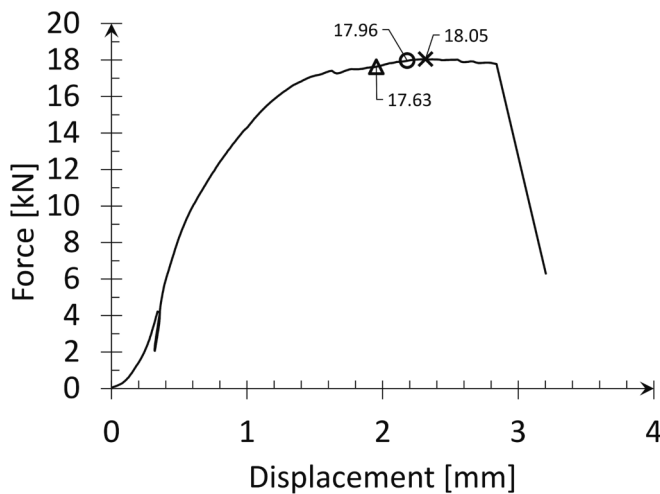


Fig. 6. Comparison of different embedment values for a Radiata pine FH sample with $d = 16$ mm, derived with 5%-offset between 10% and 40% ($F_h = 17.63$ kN), 5%-offset between 20% and 60% ($F_h = 17.96$ kN), and maximum force within first 5 mm after reloading point ($F_h = 18.05$ kN).

$$f_{h,5\%,m} = f_{h,5mm,m} = 0.08757\rho_m \quad (7)$$

Leijten and Köhler [15]:

$$\text{Softwoods : } f_{h,5mm,m} = 0.097\rho_m^{1.07}d^{-0.25} \quad (8.1)$$

$$\text{Hardwoods : } f_{h,5mm,m} = 0.087\rho_m^{1.09}d^{-0.25} \quad (8.2)$$

Traetta and Schickhofer [61] (FH C):

$$f_{h,5mm,k} = 0.1 d^{-0.15}\rho_{12,k} \quad (9)$$

Sandhaas et al. [20] (FH C):

$$f_{h,5mm,k} = 0.082\rho_k \quad (10)$$

This preliminary analysis partly meets the second objective of this study, i.e. to assess the applicability of embedment formulae proposed in literature to Australian softwoods. The results for Mack's data are shown in Table 2, and results for the present experimental study are discussed further below. Note that no differentiation was made between characteristic and mean values in Table 2, since this would require small sample statistics such as given in Leicester [108]. Instead, mean density was inserted into prediction formulae to calculate embedment strength, regardless of whether or not these were calibrated to predict characteristic strength.

Several more strength prediction models for both 5%-offset and maximum embedment strength were selected from literature based on their relevance to the present experimental study where softwood was loaded parallel to the grain in FH tension and HH compression. The validity of strength predictions was tested by comparison of mean and characteristic strengths obtained in this study, and results are discussed further below.

Further full hole predictions for $f_{h,5mm}$ used to calculate ultimate strength are given in Equations (11) to (12.2).

Yurrita and Cabrero [73,21] (literature study):

$$f_{h,5mm,m} = (0.0095 + 0.007\rho_m/1000)\rho_m^4\sqrt{f_y} \quad (11)$$

Franke and Quenneville [18,19] (HH C) used to predict maximum strength:

$$f_{h,5mm,k} = 0.067\rho_k \quad (12.1)$$

$$f_{h,5mm,k} = 0.072(1-0.0024d)\rho_k \quad (12.2)$$

The Half Hole strength prediction for 5%-offset embedment strength parallel to grain $f_{h,5\%}$ used to calculate proportional limit strength of joints is given in Equation (13), converted from the NDS [27] to MPa.

NDS [27] (HH C) for $f_{h,5\%,m}$:

$$f_{h,5\%,m} = 0.07725\rho_m[\text{MPa}] \quad (13)$$

The third objective was to compare embedment strength obtained from FH tension and HH compression tests. This prompts a discussion of the purpose of embedment tests. Different embedment test methods have already been discussed in literature [67,71,77], concluding that FH tension tests are most suited to mimic three-member timber joints or steel-to-timber joints used in practice. It is reasonable to assume that ultimate embedment strength predictions ($f_{h,5mm}$ or $f_{h,max}$) are best suited to calculate ultimate joint capacity, designated F_u in Eurocode 5 [26]. The suitability of F_u as a design point has been challenged in literature [1,2,34], however this will not be further discussed here.

In the context of the NDS [27], proportional limit strength of joints is predicted with $f_{h,5\%}$, derived with the 5%-offset method from HH tests [1,12]. Based on Hager [38], it may be stipulated that the highest embedment stress observed in FH testing is 2.75 times the average embedment stress, or equivalent embedment stress from HH testing, i.e. $f_{h,FH,max} \approx 2.75 f_{h,HH,max}$. However, this only relates to stress concentrations and not an embedment strength derived from load displacement curves. Yet, the discussion how HH embedment strength may relate to FH embedment strength is worthwhile, since HH testing may be a viable alternative for materials such as CLT, where the embedment strength exceeds the fastener yield moment. Eliminating dowel bending in FH testing of CLT requires high strength dowels, where yield strength in turn affects embedment strength [21,71,109]. It is then possible to carry

Table 4

Selection of matched pairs of half hole (HH) and full hole (FH) samples where density ρ_{12} and MC differed no more than 5%. Radiata Pine (RP), Southern Pine (SP), dowel diameter d, equivalent density at 12% MC ρ_{12} , embedment strengths established with the 5% offset method: $f_{h,5\%(10-40)}$ established between 10% and 40% of F_{max} and $f_{h,5\%(20-60)}$ established between 20% and 60% of F_{max} . Maximum embedment strength within 5 mm after reloading $f_{h,5mm}$. Ratios between HH and FH properties printed in bold.

Species	Board	Method	d [mm]	ρ_{12} [kg/m ³]	MC [%]	$f_{h,5mm}$ [MPa]	$f_{h,5\%(10-40)}$ [MPa]	$f_{h,5\%(20-60)}$ [MPa]
RP	8	HH	12.7	390.4	11.3%	30.74	30.05	30.21
RP	11	FH	12.7	412.9	11.6%	31.30	30.04	29.83
				0.95	0.97	0.98	1.00	1.01
RP	2	HH	12.7	431.0	11.1%	33.31	31.37	32.08
RP	11	FH	12.7	426.2	11.1%	34.66	32.63	33.71
				1.01	1.00	0.96	0.96	0.95
RP	4	HH	12.7	468.2	11.3%	30.67	30.12	30.05
RP	12	FH	12.7	468.3	11.8%	35.27	33.29	34.48
				1.00	0.96	0.87	0.90	0.87
RP	1	HH	12.7	464.3	12.1%	33.15	30.63	32.01
RP	9	FH	12.7	465.1	12.4%	36.73	33.79	35.12
				1.00	0.98	0.90	0.91	0.91
RP	1	HH	12.7	457.5	11.7%	33.10	25.95	31.52
RP	12	FH	12.7	458.6	11.5%	34.96	32.64	33.58
				1.00	1.03	0.95	0.79	0.94
RP	1	HH	12.7	459.3	12.0%	34.55	32.18	33.81
RP	7	FH	12.7	458.9	12.0%	36.87	33.43	34.45
				1.00	1.01	0.94	0.96	0.98
RP	1	HH	12.7	463.8	12.4%	31.05	29.39	29.70
RP	9	FH	12.7	462.0	12.5%	34.31	29.33	32.14
				1.00	1.00	0.91	1.00	0.92
RP	15	HH	16	373.2	11.2%	23.08	22.42	22.46
RP	10	FH	16	380.3	11.6%	25.99	25.58	25.61
				0.98	0.96	0.89	0.88	0.88
RP	18	HH	16	437.9	11.9%	27.34	26.72	27.00
RP	11	FH	16	439.3	11.7%	32.42	32.03	32.01
				1.00	1.02	0.84	0.83	0.84
RP	18	HH	16	447.5	11.8%	26.20	25.97	26.13
RP	3	FH	16	448.0	12.2%	29.69	29.47	29.48
				1.00	0.96	0.88	0.88	0.89
RP	16	HH	16	458.3	11.4%	29.38	28.63	28.72
RP	9	FH	16	459.0	11.9%	29.46	29.38	29.45
				1.00	0.96	1.00	0.97	0.98
SP	14	HH	12.7	422.6	11.6%	26.76	25.96	26.45
SP	15	FH	12.7	433.1	11.0%	28.26	26.86	27.10
				0.98	1.05	0.95	0.97	0.98
SP	14	HH	12.7	448.2	11.4%	26.98	23.41	25.49
SP	6	FH	12.7	452.4	11.4%	30.50	28.95	29.71
				0.99	1.00	0.88	0.81	0.86
SP	3	HH	16	502.0	11.4%	34.65	34.13	34.59
SP	19	FH	16	496.5	11.8%	33.90	33.90	33.90
				1.01	0.96	1.02	1.01	1.02
SP	2	HH	12.7	490.8	13.1%	39.40	39.40	39.40
SP	3	FH	12.7	488.3	13.4%	35.77	33.49	34.18
				1.01	0.98	1.10	1.18	1.15
				mean		0.94	0.94	0.95

out embedment tests without distortion of results from using an incorrect dowel material in a typical material thickness configuration for materials such as Glulam, LVL or CLT [110]. The relationship between $f_{h,5\%}$ and $f_{h,5mm}$ derived from matched HH and FH specimens is discussed further below.

4. Results and discussion

Fig. 5 shows the load displacement curves for each embedment test setup. The curves were aligned at the reloading point in order to obtain more homogenous results. The experimental embedment strength was then established on each load-displacement curve with both the 5% offset method ($f_{h,5\%}$) and within the first 5 mm of displacement ($f_{h,5mm}$) after the reloading point.

Table 3 gives the experimental characteristic embedment strengths for each species and setup. Characteristic density and embedment strengths were determined as the 5-th percentile value estimated with 75% confidence according to AS/NZS 4063.2:2010 [105] Method 4, which is the same as ASTM D2915-17 [111] and valid for $n = 28$. The 5%-offset values were established between 10% and 40% of F_{max} as

suggested by Sawata and Yasumura [16] and Franke and Magnière [67], as well as 20% and 60% since this represents the stiffness after the reloading point.

4.1. Relationship between density, fastener diameter, and embedment strength

A significant positive correlation between timber density ρ and embedment strengths was found for both FH and HH tests, regardless of how embedment strength was defined (95% confidence, $\alpha = 0.05$, one-tailed student t -test). However, only the FH test data showed a significant negative correlation between fastener diameter and embedment strength, for both maximum embedment strength $f_{h,5mm}$ and 5%-offset strength after the reloading point $f_{h,5\%(20-60)}$ (95% confidence, $\alpha = 0.05$, one-tailed student t -test). For HH tests, fastener diameter and embedment strength were not significantly correlated. This is consistent with previous findings by Wilkinson [12], and the formula contained in the NDS [27].

Table 5

Strength predictions according to literature for mean and characteristic embedment strength based on mean and characteristic experimental density, and respective experimental strength values. c = conservative, nc = non-conservative. Note that Equation 3.2 might be a decimal print error in Mack [44].

Mean proportional or 5% offset strength	FH Radiata Pine		FH Southern Pine		
d [mm]	12.7	16	12.7	16	
ρ_m [kg/m ³]	438.1	429.7	561.3	547.8	
$f_{h,5\%(10-40),m}$ [MPa]	29.5	28.9	41.7	37.0	
$f_{h,5\%(20-60),m}$ [MPa]	31.4	29.0	42.8	37.1	
[44] (FH C) Equ. (3.2)	1.8	n/a	2.1	n/a	c*
$f_{h,y,m} = 0.129 \rho_{12}^{0.68} / (dt) * 1000$					
[16] (FH C) Equ. (7)	38.4	37.6	49.2	48.0	nc
$f_{h,5\%,m} = 0.08757 \rho_m$					
Mean maximum strength	FH Radiata Pine		FH Southern Pine		
d [mm]	12.7	16	12.7	16	
ρ_m [kg/m ³]	438.1	429.7	561.3	547.8	
f_y [MPa]	665	665	665	665	
$f_{h,5mm,m}$ [MPa]	33	29.4	44.6	37.7	
[44] (FH C) Equ. (3.1)	30.0	n/a	39.7	n/a	c
$f_{h,max,m} = 0.0138 \rho_{12}^{1.13} / (dt) * 1000$					
[16] (FH C) Equ. (7)	38.4	37.6	49.2	48.0	nc
$f_{h,5mm,k} = 0.08757 \rho_m$					
[56] Equ. (8.1)	34.5	31.9	44.9	41.3	nc
$f_{h,5mm,m} = 0.097 \rho_m^{1.07} d^{-0.25}$					
[21] Equ. (11)	28.0	27.3	38.3	37.1	c
$f_{h,0,m} = (0.0095 + 0.007 \rho_m / 1000)$					
$\rho_m^{1/4} \sqrt{f_y}$					
Characteristic maximum strength	FH Radiata Pine		FH Southern Pine		
d [mm]	12.7	16	12.7	16	
ρ_k [kg/m ³]	366.5	367.4	433.2	420.6	
$f_{h,5mm,k}$ [MPa]	25.5	23.9	28.3	24.3	
[33] (FH T) Equ. (4)	22.0	21.1	26.1	24.2	c
$f_{h,5mm,k} = 0.1 \rho_k d^{-0.2}$					
Eurocode 5 [26] Equ. (5)	26.2	25.3	31.0	29.0	nc
$f_{h,5mm,k} = 0.082 (1 - 0.01 d) \rho_k$					
[61] (FH C) Equ. (9)	25.0	24.2	29.6	27.7	nc
$f_{h,5mm,k} = 0.1 d^{-0.15} \rho_{12,k}$					
[20] (FH C) Equ. (10)	30.1	30.1	35.5	34.5	nc
$f_{h,5mm,k} = 0.082 \rho_k$					
AS 1720.1 (2010) [31].	28.1	27.9	35.7	35.4	nc
$f_{h,max,k}$ for d = 12,16 mm					
Mean 5% offset strength	HH Radiata pine		HH Southern pine		
d [mm]	12.7	16	12.7	16	
ρ_m [kg/m ³]	464.9	463.8	508	511.8	
$f_{h,5\%(10-40),m}$ [MPa]	25.3	23.7	26.0	23.8	nc
$f_{h,5\%(20-60),m}$ [MPa]	25.5	23.8	27.2	24.0	nc
NDS [27] (HH C) (13)	35.9	35.8	39.2	39.5	nc
$f_{h,5\%,m} = 0.07725 \rho_m$					
Characteristic maximum strength	HH Radiata Pine		HH Southern Pine		
d [mm]	12.7	16	12.7	16	
ρ_k [kg/m ³]	390.5	373.3	418.0	442.3	
$f_{h,5mm,k}$ [MPa]	28.7	23.1	25.3	28.4	
[18] Equ. (12.1)	26.2	25.0	28.0	29.6	nc
$f_{h,5mm,k} = 0.067 \rho_k$					
[19] Equ. (12.2)	27.3	25.8	29.2	30.6	nc
$f_{h,5mm,k} = 0.072 (1 - 0.0024 d) \rho_k$					

4.2. Difference between 5%-offset and maximum strength

Depending on how 5%-offset values are established, $f_{h,5\%}$ and $f_{h,5mm}$ become almost identical, as illustrated in Fig. 6. Sawata and Yasumura [16] and Franke and Magnière [67] stated that both values were identical for tests parallel to the grain, where 5%-offset was taken between 10% and 40% of F_{max} . A similar trend was observed in the present study where the mean $f_{h,5\%(10-40)}/f_{h,5mm}$ ratio and standard deviation (SD) were 0.95 (0.06) for FH and 0.97 (0.04) for HH tests, despite FH and HH samples not being perfectly matched. When offset was established between 20% and 60% of F_{max} , the difference was even less pronounced, with mean $f_{h,5\%(20-60)}/f_{h,5mm}$ ratios and standard deviation (SD) of 0.97

(0.03) for FH test, and 0.98 (0.02) for HH tests. This somewhat contradicts the observation by Smith and Foliente [1], who stated that 5%-offset gives a value between the proportional limit and ultimate (maximum) strength.

4.3. Difference between FH and HH results

Hager [38] stated that the peak embedment stress observed at the edge of the timber in FH tests is 2.75 times that of a uniform stress distribution, as would be achieved in HH tests or by averaging embedment stress over the thickness of the sample. Furthermore, Franke and Magnière [67] found that HH specimens displayed 30% higher embedment strength than FH specimens, however, their sample size was small ($n = 5$).

In the present study, little difference was found between matched HH and FH samples for $f_{h,5\%(10-40)}$, $f_{h,5\%(20-60)}$, and $f_{h,5mm}$; on average, HH strengths were 6% lower than respective FH strengths as shown in Table 4. Hence, no strength conversion is needed for embedment testing of sawn softwood parallel to the grain if both FH and HH testing follow preloading and load displacement curves are aligned at the reloading point.

However, this may not be the case for embedment perpendicular to the grain, or if HH tests are carried out without preloading, or if load displacement curves aren't aligned at reloading. The issue should be further investigated for loading at different angles to the grain, as well as hardwoods and engineered wood products, such as LVL, Glulam and CLT.

4.4. Difference between experimental results and embedment strength formulae

Table 5 gives the strength predictions based on characteristic and mean densities using selected embedment formulae from literature, those contained in Eurocode 5 [26], NDS [27], and the respective joint group strength values in AS 1649–2001 [112] and AS 1720.1–2010 [31], which are based on a mean density range. In order to use the strength prediction by Yurrita and Cabrero [21], the yield strength of the dowels needs to be determined. AS 1444–2007 [107] suggests using the 0.2% Proof Stress to represent the yield stress, which gives $f_y = 665$ MPa. Embedment strength prediction with AS 1720.1–2010 [31] can be done indirectly, by deducing the embedment strength from the respective joint group. According to AS 1720.1–2010 [31] Table H2.4, Radiata Pine can be conservatively classified as joint group JD5 since it was originally graded MGP10, and Southern Pine is conservatively classified as joint group JD4 (mixed Australian grown pine species). For an effective thickness of $t = 35$ mm, this yields the following characteristic capacities (per shear plane) and characteristic embedment strengths for JD5: 5.9 kN and $f_{h,k} = 28.1$ MPa for $d = 12$ mm, and 7.8 kN and $f_{h,k} = 27.9$ MPa for $d = 16$ mm. For JD4 the respective characteristic capacities and embedment strengths are: 7.5 kN and $f_{h,k} = 35.7$ MPa for $d = 12$ mm, and 9.9 kN and $f_{h,k} = 35.4$ MPa for $d = 16$ mm.

The first remarkable observation is that all strength predictions in design standards, i.e. Eurocode 5 [26], AS 1720.1–2010 [31], and NDS [27], were non-conservative. However, those models specifically developed for Australian timber, such as Mack [44] and Smith et al. [33], were conservative for softwoods in Mack's study from 1981, as well as the present experimental study.

While Traetta and Schickhofer [61] was conservative for Mack's results [44], it was non-conservative for the present study. Interestingly, Yurrita & Cabrero [21] was conservative for the present study, despite the model not taking the dowel diameter into account, while a significant negative correlation between dowel diameter and embedment strength was established from experimental data for FH specimens.

Arguably, only those prediction models that accurately predict characteristic embedment strength are useful in the context of present-day design standards. However, not all prediction models were

Table A1

Dowel embedment tests in literature and proposed embedment strength formulae, with $f_{h,\alpha}$ = embedment strength in MPa at angle α relative to grain, ρ_{MC} = density in kg/m³ at given moisture content (MC in %), d = dowel diameter in mm, and t = specimen thickness in mm, unless indicated otherwise. The subscript k indicates characteristic (5%-percentile) values, and the subscript m indicates mean values. FH = full hole, HH = half hole, T = tension, C = compression.

Authors	Year	Configurations	Test standard	Proposed embedment formula
Hager [38]	1930	established formula to describe non-uniform embedment stress distribution under a dowel	n/a	peak embedment stress (FH): $\sigma_{h,0,FH(peak)} = 2.75F/(dt)$ uniform embedment stress (HH) : $\sigma_{h,0,HH} = F/(dt) = \sigma_{h,0,FH} / 2.75$ $f_{h,0} = 0.64 f_{c,0}$... for softwoods $f_{h,0} = 0.80 f_{c,0}$... for hardwoods $f_{h,90} = 1.68 f_{c,90}$
Trayer [39]	1932	embedment of bolts in Douglas Fir, Southern Yellow Pine, Sitka Spruce, Oak, and Maple, different t/d ratios, $\alpha = 0^\circ, 90^\circ$ FH tension with $t/d \leq 2$ is considered equivalent to HH embedment test	HH C, parallel and perp to grain	$f_{h,t} = 0.8 f_{h,c}$ original formulae: $\sigma_{h,0} = 1500 \gamma - 450$ [σ in kg/cm ² and γ in g/cm ³] $\sigma_{h,90} = 1500 \gamma - 800$ [σ in kg/cm ² and γ in g/cm ³] converted with 1 kg/cm ² = 0.098 MPa: $f_{h,0} = 0.147 \rho - 44.15$; $f_{h,90} = 0.147 \rho - 78.48$ or expressed as a function of the experimental embedment strength of a $d = 10$ mm dowel $f_{h,0,d10}$ $f_{h,0} = f_{h,0,d10} (0.9 + 1/d)$... see [21] $f_{h,MC} = (26 f_{h,12}) / (MC + 14)$ $f_{h,0} = 0.6 f_{c,0} + 6$... according [14]
Fahlbusch [9,40]	1949, 1951	Pine, Linden, Ash, $d = 10, 12$ mm, MC = 5–30%, $t/d < 2$	FH, T ($f_{h,t}$) and C ($f_{h,c}$), f_h determined with 2%-offset method	
Kuipers and Vermeyden [41]	1965	564 matches pairs of European Spruce, $d = 6.7$ mm, while technically nail embedment, the diameter is rather large. MC = 12% as described in [14]	n/a	
Larsen and Sorensen [42]	1973	Nordic Spruce and Fir as described in [14]	n/a	$f_{h,0} = 0.7 f_{c,0}$... according to [14]
Norén [43]	1974	As presented in [21]	n/a	$f_{h,0} = f_{h,0,d10} (66-d)/56$... see [21]
Mack [44]	1981	Effectively FH compression test (one bolt in timber with two steel side members). Seasoned Australian softwoods and hardwoods: Radiata Pine, Hoop Pine, Loblolly Pine, Slash Pine, Caribbean Pine, Patula Pine, Spotted Gum.	FH C, parallel to grain.	$F_{max,0} = 0.046 t d$ (Radiata pine) $F_{y,0} = 0.015 t d$ (Radiata pine) $F_{max,0} = 0.0138 \rho^{1.23}$ (for $d = 12$ mm, $t = 35$ mm) $F_{max,0} = 0.366 F_c$ F_c = crushing (compression) strength of timber adjacent to bolt on 20x20x60 mm samples
Whale and Smith [10]	1986	840 tests on five European softwoods (UK grown Sitka Spruce, Scots Pine, European Redwood and	FH T and C parallel to grain. FH C perp to grain. 9 preloading cycles. Basis for EN 383 standard and Eurocode 5 embedment strength formula, $f_{h,5mm}$	Whale and Smith [10] Strength properties of tensile and compressive specimens parallel to grain very similar. $f_{h,0} = (0.082 - 0.00082 d) \rho_{12}$... for softwoods and tropical hardwoods $f_{h,90} = (0.041 - 0.00088 d) \rho_{12}$... for softwoods $f_{h,90} = (0.043 - 0.00081 d) \rho_{12}$... for tropical hardwoods $f_{h,\alpha,k} = [0.0008 \rho_k (100-d)] / (2.3 \sin^2 \alpha + \cos^2 \alpha)$ Whale et al. [45] $f_{h,d} = (0.1 - 0.0016 d) \rho / (2.5 \sin^2 \alpha + \cos^2 \alpha)$... for softwoods $f_{h,d} = (0.1 - 0.0016 d) \rho / (2.2 \sin^2 \alpha + \cos^2 \alpha)$... for hardwoods Tension and compression embedment specimens exhibit different embedment characteristics. Smith et al. [46] also compared these values to other proposed embedment formula at the time. $f_{h,0} = 91.44 G - 11.16$ $f_{h,90} = (25.04 G + 1.35) (d_h/10)^{-0.4}$ d_h = bolt hole diameter in mm original formulae where D is in inches $f_{c,0} = 11,830 G_{12}$ [psi]; $f_{c,90} = 6,620 G_{12}^{1.45} D^{-0.5}$ [psi] $G_{12} = G_{dry} / (1 + 0.108 G_{dry})$, where G_{12} = specific gravity at 12% MC, G_{dry} = oven-dry specific density conversion to MPa with 1 psi = 6.89×10^{-3} MPa, $D = d/25.4$ mm, and $G_{12} = \rho_{12} / 1000$ $f_{h,0} = 0.0815 \rho_{12}$ [MPa] $f_{h,90} = 6.89 \times 10^{-3} (0.2957 \rho_{12}^{1.45}) (d/25.4)^{-0.5}$
Hirai [11]	1989	As described in [47] tests on Spruce, Hemlock, Douglas Fir with $\rho = 380 \dots 550$ kg/m ³	n/a	$= 0.01 \rho_{12}^{1.45} d^{-0.5}$ [MPa] $f_{h,MC} = 46.7 - 1.35 MC$ (for $MC < 22.5\%$) $f_{h,MC} = 16.5$ for MC greater than 22.5%
Wilkinson [12]	1991	Total of 240 bolt specimens. Southern Pine with $D = \frac{1}{2}, \frac{1}{4}, 1, 1.5$ in (12.7–41.3 mm) Douglas Fir, Southern Pine, Spruce-Pine-Fir, Sitka Spruce, Red Oak, Yellow Poplar, Aspen, with $D = 3/4$ in (19.1 mm) Thickness $t = 0.49$ – 2.25 in (12.4 – 57.2 mm)	HH C, basis for ASTM D5764 [48] and NDS 5% offset or maximum load	
Koponen [49]	1991	Finnish Spruce, $d = 12, 20$ mm, MC = 5% - “green”, as described in [14]. (The MC of green timber can range from the FSP to 100%).	n/a	
Larsen and Riberholt [50]	1991	as described by Pedersen [51]	n/a	$f_{h,\alpha} = 0.67 f_{c,0} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$ $k_{90} = 0.45 + 8d^{-1.5}$
	1992	125 compression and tension parallel to grain on European Beech, European Oak, Teak, Merbau,	prEN 383:1990 [53], basis for Eurocode 5, FH C and T parallel to grain, FH C	regression analysis softwoods:

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Table A1 (continued)

Authors	Year	Configurations	Test standard	Proposed embedment formula
Ehlbeck and Werner [12,52]		Afzelia, Azobe, d = 8,12,20,30 mm 90 tests on Beech and Azobe $\alpha = 30^\circ, 45^\circ, 60^\circ, 90^\circ$ d = 8,16,30 mm 80 tests on European Whitewood perp to grain	perp to grain $f_{h,5mm}$	$f_{h,0,k} = 0.082 (1-0.01 d) \rho$; $r = 0.49$ for 360 tests $f_{h,90,k} = 0.058 (1-0.015 d) \rho$; $r = 0.85$ for 78 tests hardwoods: $f_{h,0,k} = 0.102 (1-0.01 d) \rho$; $r = 0.68$ for 119 tests $f_{h,90,k} = 0.102 (1-0.016 d) \rho$; $r = 0.87$ for 29 tests proposal for Eurocode 5 assuming Gaussian distribution: $f_{h,0,k} = 0.082 (1-0.01 d) \rho$ $f_{h,\alpha,k} = f_{h,0,k} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$; $k_{90} = f_{h,0,k} / f_{h,90,k}$ $k_{90} = 1.35 + 0.015 d \dots$ for softwoods $k_{90} = 0.90 + 0.015 d \dots$ for hardwoods Comment: $f_{h,0}$ conservative for hardwoods! Low correlation for EC5 formula! $f_{h,0,MC} = 49.95 - 1.186 MC$
Rammer and Winistorfer [14]	2001	Total of 200 tests on Southern Pine. Additional tests on Douglas Fir-Larch, Spruce-Pine-Fir from different study MC = 4%, 6%, 12%, 19% t = 20.3 mm, d = 12.7 mm	ASTM D5764 [54]. HH C parallel to grain different MCs	
Sawata and Yasumura [16]	2002	Total of 1009 tests on different grades of Japanese Ezomatsu (Ezo Spruce) and Todomatsu (Abies Fir). d = 8,12,16,20 mm.	EN383:1993 [55] FH C parallel and perp to grain without preloading	$f_{h,0,5\%,m} = f_{h,0,5mm,m} = 0.9 f_{c,0} = 0.08757 \rho$ $f_{h,90,5\%} = 0.4 f_{c,0}$ $f_{h,90,5mm} = (0.745 - 0.016 d) f_{c,0}$ where $f_{c,0}$ is the compressive strength parallel to the grain in MPa. $f_{h,90,5\%} / f_{h,90,5mm} = 0.016 d + 0.555$ $f_{h,0,m} = 0.097 \rho^{1.07} d^{0.25} \dots$ for softwoods $f_{h,0,m} = 0.087 \rho^{1.09} d^{0.25} \dots$ for hardwoods
Leijten and Köhler [15] Leijten et al. [56]	2004	Statistical analysis of datasets by Whale and Smith [57,58], Ehlbeck and Werner [13], Sawata and Yasumura [16], unpublished data by Mischler, and Vreeswijk [59] as part of COST Action E24.	n/a	
Eurocode 5 [26]	2004		Eurocode 5 [26]	$f_{h,0,k} = 0.082 (1-0.01 d) \rho_k$ $f_{h,\alpha,k} = f_{h,0,k} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$ $k_{90} = 1.35 + 0.015 d \dots$ for softwoods $k_{90} = 0.90 + 0.015 d \dots$ for hardwoods $f_{h,0,k} = 0.1 \rho_k d^{0.2}$ $f_{h,90,k} = 1.9 \rho_k^{0.6} d^{0.5}$
Smith et al. [33] Foliente et al. [35]	2004	total of 60 embedment tests on different pine grades (JD), 10 replicates each: Radiata pine (JD4) d = 8,16 mm, $\alpha = 0^\circ$, Radiata Pine (JD5) d = 8,16 mm, $\alpha = 0^\circ$, Radiata Pine (JD5) d = 16 mm, $\alpha = 90^\circ$, and Slash Pine (JD3) , d = 8, $\alpha = 0^\circ$ JD3: $\rho = 600 \dots 745 \text{ kg/m}^3$, $\rho_k = 500 \text{ kg/m}^3$ JD4: $\rho = 480 \dots 595 \text{ kg/m}^3$, $\rho_k = 400 \text{ kg/m}^3$ JD5: $\rho = 380 \dots 475 \text{ kg/m}^3$, $\rho_k = 320 \text{ kg/m}^3$ MC = 8.2%...13.5%	EN 383:1993 [55], FH T parallel to grain, FH C perp to grain	
Jumaat et al. [60]	2006	Embedment tests on Malaysian hardwood: Kempas, Mengkulang, Pulau. d = 6,8,12 mm	EN 383:1993 [55] FH C	$f_h = 0.103 (1-0.013d) \rho$
Traetta and Schickhofer [61]	2007	total of 800 tests on two softwoods: Spruce (C24) d = 8, 12, 20 mm Pine $\rho = 480-550 \text{ kg/m}^3$ and $580-600 \text{ kg/m}^3$ d = 12 mm t = 2d or 1.75d $\alpha = 0^\circ, 10^\circ, 20^\circ, 25^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ, 70^\circ, 80^\circ, 90^\circ$ for d = 12 mm Spruce; $\alpha = 0^\circ, 90^\circ$ for all other configurations	prEN 383:2006 [62] FH C with preloading	$f_{h,0,k} = 0.1 d^{-0.15} \rho_{12,k}$ $f_{h,90,k} = 0.1 d^{-0.32} \rho_{12,k}$ with $\rho_{12,k} = \rho_{MC,k} [1 + 0.005 (12 - MC)]$ $f_{h,\alpha,k} = f_{h,0,k} / [(1 - k_{90}) \cos^4 \alpha + k_{90}]$ with $k_{90} = d^{0.17}$
Hübner and Schickhofer [63]	2008	total of 776 tests on Ash, $\rho_{mean} = 751 \text{ kg/m}^3$ d = 6,8,12,16,20 mm; $\alpha = 0^\circ, 30^\circ, 60^\circ, 90^\circ$ and $\alpha = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ$ for d = 12 mm, t/d ≤ 2	prEN 383:2006 [62], FH C	$f_{h,0} = 0.1003 (1 - 0.0155 d) \rho$; $f_{h,\alpha} = f_{h,0} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$ $k_{90} = 0.5902 + 0.0329 d \dots$ for $d \geq 6 \text{ mm}$ $f_{h,0,k} = 0.08797 (1 - 0.0155 d) \rho_k$
Hübner et al. [17]	2008	total of 2187 tests on European hardwoods: Beech, Ash, Robinia. Mean density of test sets at 12% MC $\rho_m = 681 \dots 0.818 \text{ kg/m}^3$ d = 6,8,12,16,20 mm; $\alpha = 0^\circ, 30^\circ, 60^\circ, 90^\circ$ and $\alpha = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ$ for ash d = 12 mm; t/d ≤ 2	EN 383:2007 [64], FH, C and T	$f_{h,0} = 94.233 - 2.543 MC$; $f_{h,90} = 83.281 - 1.851 MC$ $f_{h,\alpha,k} = f_{h,0,k} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$ $f_{h,0,k} = 0.015 \rho_k^{1.3} d^{-0.2} \dots$ for European hardwoods $k_{90} = 0.62 + 0.035 d \dots$ for European hardwoods
Franke and Quenneville [18]	2009	188 tests on New Zealand Radiata Pine, d = 10,16,20 mm; $\alpha = 0^\circ, 90^\circ$	ASTM D5764-97a [65] HH, C, 5%-offset	$f_{h,0,k} = 0.067 \rho_k \dots$ for NZ radiata pine $f_{h,\alpha,k} = f_{h,0,k} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$ $k_{90} = 2.0 \dots$ for NZ radiata pine Comment: used ASTM HH but compared results to EC5 formula
Franke and Quenneville [19]	2011	New Zealand Radiata Pine	ASTM D5764-97a [65] HH, C, 5%-offset	$f_{h,0,k} = 0.072 (1-0.0024 d) \rho_k \dots$ for NZ radiata pine $f_{h,\alpha,k} = f_{h,0,k} / (k_{90} \sin^2 \alpha + \cos^2 \alpha)$ $k_{90} = 2.0 \dots$ for NZ radiata pine Comment: used ASTM HH but compared results to EC5 formula

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Table A1 (continued)

Authors	Year	Configurations	Test standard	Proposed embedment formula
Glisovic and Stevanovic [66]	2012	50 tests on Pine and Oak, $d = 12$ mm, $t/d = 2.5$, $\alpha = 0^\circ$	EN 383:1993 [55] FH C	EC5 underestimates hardwood embedment strength by 15% (as expected)
Sandhaas et al. [20]	2013	139 tests on European Spruce, European Beech, Purpleheart, Cumaru Peru, Cumaru Brazil, Azobe; $d = 12, 24$ mm; $\alpha = 0^\circ$; $t/d = 1.67$; high strength steel $f_u = 590$ MPa and very high strength steel $f_u = 1389$ MPa	EN 383:2007 [64] FH, C parallel to grain	$f_{h,0,k} = 0.082 \rho_k$ steel grade influences embedment strength
Franke and Magnière [67]	2014	30 tests on Spruce, $d = 12$ mm (mild steel), $\alpha = 0^\circ$, 90° , 5 replicas per method (limited statistical significance), MC = 12% (20 °C, 65% RH)	EN 383:2007 [64] FH C and ASTM D5764-97a [68] HH C, FH C, both with preloading, $f_{h,5mm}$ and $f_{h,5\%}$	recommendations for test method: use ASTM HH (presumably with preloading) to obtain $f_{h,5\%}$ as embedment strength use EN 383 to derive stiffness (foundation modulus) specimen size according to EN 383 with the additional requirement of $t \geq \min(40 \text{ mm}, 4d)$.
NDS [27]	2018	slight variation from Wilkinson [12] where $f_{e,0} = 11,830 G_{12} [\text{psi}]$ $f_{e,90} = 6,620 G_{12}^{1.45} D^{-0.5} [\text{psi}]$ and converted to MPa $f_{h,0} = 0.0815 \rho_{12} [\text{MPa}]$ $f_{h,90} = 0.01 \rho_{12}^{1.45} d^{-0.5} [\text{MPa}]$	NDS [27]	$f_{e,0} = 11200 G$ and $f_{e,90} = 6100 G^{1.45} D^{-0.5}$ where f_e = embedment strength in psi (lb/in ²), with specific gravity G and fastener diameter D in inches. Conversion to MPa with 1 psi = 6.89×10^{-3} MPa, 1 in = 25.4 mm, $G = \rho/1000$. $f_{h,0} = 0.07725 \rho [\text{MPa}]$ $f_{h,90} = 0.0095 \rho^{1.45} d^{-0.5} [\text{MPa}]$ Comment: These are predictions are for the proportional limit ($f_{h,5\%}$). $f_{h,\alpha} = f_{h,0} f_{h,90} / (f_{h,0} \sin^2 \alpha + f_{h,90} \cos^2 \alpha)$ $f_{h,0} = 50 G (1 - 0.01 d)$ $f_{h,90} = 22 G (1 - 0.01 d)$ Comment: These predictions are for the maximum embedment capacity ($f_{h,max}$).
CSA O86 [28]	2014	based on tests on bolts (Smith et al. [46]; Whale and Smith [23,69,70]); no product test standard available	CSA O86 [28]	
Molina et al. [71]	2017	48 embedment tests on Slash Pine and Blue Gum, $d = 6.35$ mm (1/4 in), $\alpha = 0^\circ, 90^\circ$ according to ABNT NBR 7190:1997 [29] and EN 383:2007 [64]	FH C, parallel and perp to grain, ABNT NBR 7190:1997 [29] with 2‰ method $f_{h,2\%}$ EN 383:200 [64], $f_{h,5mm}$	ABNT NBR 7190:1997 [29]: $f_{h,0} = f_{e,0}$; $f_{h,90} = 0.25 f_{e,0} \alpha_e$ found to be non-conservative; recommendation to use EN 383 [64]. $f_{h,2\%} < f_{h,5mm}$
Malek et al. [72]	2018	19 embedment tests on Mengkulang, Kempas, $d = 12$ mm, $t = 50$ mm, $\alpha = 0^\circ$	HH C, parallel to grain, ASTM D5764-97a [63] with $f_{h,5\%}$ and $f_{h,5mm}$	$f_{h,5mm} > f_{h,5\%}$ (as expected)
Yurrita and Cabrero [73,21]	2017, 2018	New proposed formula that considers steel yield strength f_y . Proposal based on Sandhaas et al. [20] dataset and validation with Leijten et al. [56] dataset.	n/a	$f_{h,0,m} = (0.0095 + 0.007 \rho_m / 1000) \rho_m^4 / f_y$
Schweigler et al. [74]	2019	developed a beam on elastic foundation embedment model to describe FH embedment load–displacement curves based on 7 FH datasets from ENSTIB/LERMaB Epinal, Vienna University of Technology (IMWS, TU Wien), Linnaeus University Växjö (LNU), TU Delft and Karlsruhe Institute of Technology (KIT).	EN 383:2007 [64] FH	recommended to increase displacement to 15 mm

developed for characteristic strength, and Yurrita and Cabrero [21] pointed out that characteristic formula may be used with mean densities to predict mean embedment strength values, but not vice versa. This statement is supported by the present study. If mean prediction models are applied to characteristic values, Mack [44] becomes non-conservative for Southern Pine, and Yurrita and Cabrero [21] becomes non-conservative for Southern Pine with $d = 16$ mm. Both Sawata and Yasumura [16] and Leijten & Köhler [55] remain non-conservative.

From Wilkinson [12] and the NDS [27] it is not entirely clear whether the models are intended for the prediction of characteristic or mean embedment strength. Regardless, their predictions are non-conservative for both characteristic and mean HH embedment strength in the present study.

5. Conclusions

This paper reviewed and discussed the development of international embedment strength prediction formulae since 1930. Furthermore, an experimental study comprising of a total of 224 full hole (FH) tension and half hole (HH) compression embedment tests were conducted parallel to the grain according to AS/NZS ISO 10984.2:2015 [93].

The main objectives were to compare embedment strength obtained

FH tension and HH tests, establish embedment strength values for two commonly used Australian softwoods, and to assess the applicability of embedment formulae proposed in literature.

In order to compare embedment strength resulting different test methods, FH specimens were sized to EN 383:2007 [64], and HH specimens were sized to ASTM D5764-97a [103], respectively. Two different dowel diameters (12.7 mm and 16 mm) and two Australian sawn softwoods (Radiata Pine and Southern Pine) were employed with 28 replicates per setup (224 samples in total). All samples were subjected to monotonic loading with preloading to 20% of the expected maximum strength to minimise the influence of initial slip, and the embedment strength was derived from load-displacement curves with the 5%-offset method between 10% and 40% as well as 20% and 60% of the maximum force, and from the maximum force within the first 5 mm of displacement after the reloading point.

Little difference was found between the different methods used to establish embedment strength from load-displacement curves. Furthermore, little difference was found between FH and HH strengths established from density matched samples. Both FH and HH tests showed that density and embedment strengths were positively correlated. However, only FH tests showed a significant negative correlation between dowel diameter and embedment strength, and this was only the case for

embedment strength established as a 5%-offset after reloading between 20% and 60%, or as the maximum strength within the first 5 mm of displacement after the reloading point. This is consistent with the embedment strength prediction methods contained in Eurocode 5 [26] and the NDS [27], and the respective suggested experimental methods in EN 383:2007 [62] and ASTM D5764-97a [103].

Embedment strength prediction formulae in Eurocode 5 [26] and the NDS [27], as well as empirical tabulated strength data contained in AS 1720.1–2010 [31], yielded non-conservative embedment strength predictions. However, Smith et al. [33] provided a good strength prediction for Australian softwood, both in the present study and for a previous study by Mack [44].

It is therefore recommended that $f_{h,0,k} = 0.1 \rho_k d^{0.2}$ according to Smith et al. [33] be adopted in the next version of the joint proposed standard DZ NZS AS 1720.1 [81] for softwoods loaded parallel to the grain. In contrast, it is strongly recommended that the softwood formula in Eurocode 5 [26] not be adopted in the next version of DZ NZS AS 1720.1 [81] for two reasons: 1) it was non-conservative in the present study and previous study by Mack [44] and 2) it has a relatively weak correlation coefficient of $r = 0.49$.

In terms of experimental embedment strength testing, it is recommended that full hole tension tests with preloading be used to establish embedment strength, since this method is closest to the stress state encountered in an actual timber joint. Half hole compression tests are recommended where the nature of the material causes the embedment strength to exceed the dowel yield moment, as is the case for many engineered wood products, as well as hardwoods. This agrees with the current procedure given in ISO AS/NZS 10984.2:2015 [93]. Results from the present study on sawn softwoods suggest that no strength conversion is needed for embedment strength parallel to grain, however, further research on different timber materials, species, and load to grain angles is recommended.

The maximum embedment strength within 5 mm after reloading should be used to predict ultimate joint capacity and to avoid unnecessary overstrength. Since the 5%-offset strength was often similar to the maximum embedment strength, it is not suited to predict yield joint capacity. Instead, 80% of the maximum embedment strength should be used as suggested in Ottenhaus et al. (2018) [113].

Further research should investigate embedment for other Australian and New Zealand softwoods and hardwoods, and engineered wood products such as LVL, Glulam and CLT, including at different angles to the grain, in order to derive and verify reliable embedment strength formulae for the next version of DZ NSZ AS 1720.1 [81].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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